

DRAWING DESCRIPTIONS — PATENT A

Self-Sustaining Neural-Motor Energy Harvesting Loop for Autonomous Cognitive Systems

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a system architecture block diagram showing the eight-layer self-sustaining neural-motor energy harvesting loop.

FIG. 2 is a comparative energy balance chart showing control system energy depletion versus experimental system energy growth over 2000 operational steps.

FIG. 3 is a schematic diagram of the spiking neural network architecture showing the leaky integrate-and-fire neuron model with synaptic connections.

FIG. 4 is a circuit schematic of the energy harvesting subsystem comprising piezoelectric and thermoelectric transducers with rectification and storage.

FIG. 5 is a graph showing the activity-dependent energy dynamics curve illustrating the quadratic cost model and optimal activity operating region.

FIG. 6 is a hardware reference design layout diagram showing component placement and interconnections for the physical embodiment.

FIG. 7 is a validation results summary table confirming all falsifiable claims pass the specified thresholds.

FIG. 8 is a reference architecture diagram of the energy-bounded recursive control system showing the canonical signal flow from sensor interface through neuromorphic core, recursive reflection, energy governor, cognitive modulation, neural router, and actuation, with energy harvesting feedback.

DETAILED DESCRIPTION OF THE DRAWINGS

Reference Numeral Convention: Each figure in this patent uses a per-figure numbering series (FIG. 1 uses 100-series, FIG. 2 uses 200-series, through FIG. 8 uses 800-series), with even-numbered increments. Where the same physical component appears in multiple figures, it receives the numeral appropriate to that figure's series.

FIG. 1 — System Architecture Block Diagram (8-Layer Loop)

This figure shows eight labeled blocks arranged in a vertical flow with arrows indicating signal direction, forming a closed loop.

Block layout (top to bottom, then looping back):

ENVIRONMENT 100 (external, at top)

|

v

L1: Printed Membrane 102 — labeled "ThermochromicMixin"

|

v

L2: Sensing Pads 104 — labeled "light_intensity, membrane_temp"

|

v

L3: Optical Transmission 106 — labeled "signal_voltage = light / 1000 x 5.0"

|

v

L4: Neuromorphic CPU (SNN) 108 — labeled "BaseSNN (LIF neurons)"

| Arrow entering from L8: Recursive

| Reflection 116 labeled "reflection feedback"

v

L5: Servo Actuation 110 — labeled "target_angle = clip(output x 180)"

|

+-----> **L6: Energy Harvesting 112 — labeled "piezo + thermal"**

|

v

L7: Ground Reference 114 — labeled "noise floor / baseline"

|

v

L8: Recursive Reflection 116 — labeled "previous_output"

|

+-----> Recursive Feedback 120 (arrow loops back to

Neuromorphic CPU 108)

Three signal pathways should be distinguished by line style:

- Solid line: Main spine (Printed Membrane 102 through Recursive

Reflection 116 to Neuromorphic CPU 108)

- Dashed line: Thermal Cross-Link 118 (Printed Membrane 102 membrane_temp

directly to Energy Harvesting 112)

- Dotted line: Recursive Feedback 120 (Recursive Reflection 116 to

Neuromorphic CPU 108)

All blocks are rectangles with rounded corners. Environment 100 is shown as a cloud shape. The loop-back arrow (Recursive Feedback 120) from Recursive Reflection 116 to Neuromorphic CPU 108 should be prominent and labeled "THE LOOP CLOSES HERE."

FIG. 2 — Energy Balance Comparison

This figure shows a dual-panel line graph or single graph with two curves.

X-axis: Operational Steps (0 to 2000)

Y-axis: Cumulative Energy (mWh)

Experimental curve 200 (dashed line, labeled "BalancedEnergyConfig (Experimental)"):

- Starts at 50.0 mWh
- Rises rapidly to 100.0 mWh (storage capacity) within approximately 30 steps
- Plateaus at 100.0 mWh for remaining steps (homeostatic equilibrium)
- Label: "base_consumption = 45 mW, capacity = 100 mWh, overflow → heat"

Control curve 202 (solid line, labeled "EnergyConfig (Control)"):

- Starts at 50.0 mWh
- Decreases linearly
- Ends at approximately -4,697 mWh at step 2000
- Label: "base_consumption = 470 mW"

A horizontal dashed line at $y = 0$ serves as the zero energy baseline 204,

labeled "Self-Sustaining Threshold."

A horizontal dashed line at $y = 100$ serves as the capacity limit 206, labeled

"Storage Capacity (100 mWh)."

Annotations:

- Arrow pointing to control curve 202: "System fails — energy depleted"
- Arrow pointing to experimental curve 200 plateau: "System achieves homeostasis at capacity"

The area between experimental curve 200 and control curve 202 should be lightly shaded to emphasize the divergence.

FIG. 3 — SNN Architecture (LIF Neuron Model)

This figure shows the internal structure of the spiking neural network.

Top section — Single LIF Neuron Detail:

- Circle labeled Neuron i 308 (LIF model)
- Input arrows entering from left: I_external input 302 and I_reflection input 304
- Both inputs converge at summing junction 306 before entering Neuron i 308
- Inside the circle: equation written as " $V(t+1) = V(t) \times \text{leak} + I \times dt$ "
- V_th threshold 310: horizontal dashed line labeled " $V_{\text{threshold}} = 0.5$ "
- Spike output 312 arrow exiting right: "spike (0 or 1)"
- Below Neuron i 308: "If $V \geq \text{threshold}$: spike, $V = V_{\text{reset}}$ "

Bottom section — Network Topology:

- Network neurons 314: N circles (labeled "1" through "N") arranged in a single layer
- All-to-all connections shown as lines between network neurons 314
- $W[i,j]$ weight matrix 316 notation on one representative connection
- Input arrow from left entering all network neurons 314: "adjusted_input"
- spike_vector output 318 arrows from all network neurons 314 going right
- Self-connections shown as small loops, crossed out (zeroed)
- Label: "N = 50 neurons (operational), N = 20 neurons (control)"

Inset box showing STDP Learning Rule 320:

- "If post fires: $W += A+ \times \text{pre_trace}$ (LTP)"
- "If pre fires: $W -= A- \times \text{post_trace}$ (LTD)"
- " $A- > A+$ (stability bias)"

FIG. 4 — Energy Harvesting Circuit

This figure shows an electrical circuit schematic.

Left side — Piezoelectric Harvester:

- **Piezoelectric disc 400 (capacitor with "PIEZO" label)**
- **Connected to servo motor 416 shaft via mechanical coupling 406**
(zigzag line)
- **Label: "27mm piezoelectric disc"**
- **Output leads to full-bridge rectifier 402 (4 diodes in bridge configuration)**
- **Full-bridge rectifier 402 output to smoothing capacitor 404**
- **Label: "friction_factor = 18.0"**

Center — LC Resonance Circuit:

- **Ferrite inductor 418 in series with piezoelectric disc 400 output**
- **Label: "Laird ferrite-backed inductor"**
- **Resonance annotation: " $f_{\text{res}} = 1 / (2 \times \pi \times \sqrt{L \times C})$ "**
- **Label: "3-10x voltage amplification at resonance"**

Right side — Thermoelectric Harvester:

- NTC 10K 3950 thermistor 408
- Connected to thermoelectric rectifier 410
- Thermoelectric rectifier 410 output to thermoelectric smoothing capacitor 412
- Label: "thermal_factor = 8.0"

Bottom — Energy Storage:

- Both harvester outputs combine at power combiner 420
- Arrow to energy storage 414 (battery/capacitor symbol)
- Label: "energy_stored (mWh)"
- Output arrow labeled "Power to SNN (L4)"

Power budget annotation box:

- "Harvest = friction_harvest + thermal_harvest"

- " $\text{Cost} = \text{base_consumption} + \text{activity_cost} \times \text{spike_count}^2$ "

- " $\text{Net} = \text{Harvest} - \text{Cost}$ "

FIG. 5 — Activity-Dependent Energy Dynamics

This figure shows a graph derived from `BalancedEnergyConfig` parameters (core/energy.py) illustrating the energy harvesting and consumption curves as functions of neural spike count.

X-axis: "Neural Activity (Spikes / Step)" ranging from 0 to 50

Y-axis: "Energy per Step (mWh)"

Two curves:

1. Harvest curve (ref 500, dashed line): Nearly linear, gently increasing

- **Formula: $\text{Harvest} = 0.20 + (0.09 \times \text{spikes}) \text{ mWh/step}$**

- **Composed of thermal baseline (0.20 mWh at $\Delta T = 5$ degrees C) plus**

piezoelectric harvest proportional to spike-driven motor activity

- **Rises from 0.20 mWh at 0 spikes to approximately 4.7 mWh at 50 spikes**

2. Cost curve (ref 502, solid line): Quadratic, starting near harvest level

- Formula: $\text{Cost} = 0.225 + (0.006 \times s) + (0.0003 \times s^2)$ mWh/step

- Composed of base consumption ($45 \text{ mW} \times 0.005 \text{ h} = 0.225 \text{ mWh}$) plus

linear activity cost ($1.2 \text{ mW} \times 0.005 \text{ h} = 0.006 \text{ mWh/spike}$) plus

quadratic burst penalty ($0.06 \times 0.005 = 0.0003 \text{ mWh/spike squared}$)

- Rises from 0.225 mWh at 0 spikes to approximately 1.3 mWh at 50 spikes

Crossover point (ref 506): Harvest equals Cost at approximately 0.3 spikes. Below this point (0 to 0.3 spikes), cost exceeds harvest (quiescent drain). Above this point (0.3 to 50+ spikes), harvest exceeds cost (net positive).

Shaded region (ref 504, hatched): Area between curves where harvest exceeds

cost, labeled "NET POSITIVE REGION" with sublabel "Harvest >> Cost (Self-Sustaining)."

Formula annotation box (ref 508): Contains exact formulas for both curves.

Zone annotation at bottom: "Zone 1: Quiescent Drain — 0-1 Spikes: Cost > Harvest"

Zero reference line: Horizontal dotted line at $y = 0$.

FIG. 6 — Hardware Reference Design

This figure shows a top-down component layout diagram.

Central component:

- **Raspberry Pi Pico (RP2040) 600, shown as a rectangle with pin labels**

on edges

Connected components (arranged around Raspberry Pi Pico 600):

Top left:

- **Piezo Disc 27mm 602, shown as a circle**
- **Connected to Raspberry Pi Pico 600 via ADC pin (GP26/ADC0)**
- **Mechanical coupling line to SG90 Micro Servo 604**

Top right:

- **SG90 Micro Servo 604, shown as a rectangle**
- **Connected to Raspberry Pi Pico 600 via PWM pin (GP0)**

- Mechanical coupling to Piezo Disc 602 (zigzag line)

Bottom left:

- NTC 10K Thermistor 606, shown as a small component
- Connected to Raspberry Pi Pico 600 via ADC pin (GP27/ADC1)

Bottom center:

- GL5528 Photoresistor (LDR) 610, shown as a small component
- Connected to Raspberry Pi Pico 600 via ADC pin (GP28/ADC2)

Bottom right:

- WS2812B RGB LED 608, shown as a small square
- Connected to Raspberry Pi Pico 600 via digital pin (GP15)

Left side:

- Laird Ferrite Inductor 612
- Connected in series between Piezo Disc 602 and rectifier bridge 614

- Label: "LC resonance circuit"

Power section:

- Rectifier bridge 614

- Storage capacitor 616

- Arrow labeled "Harvested energy feedback"

BOM annotation box in corner:

- "Raspberry Pi Pico (RP2040)"

- "SG90 Micro Servo"

- "27mm Piezoelectric Disc"

- "NTC 10K Thermistor"

- "Photoresistor (LDR)"

- "WS2812B RGB LED"

- "Misc (wires, board, inductor)"

FIG. 7 — Validation Results Summary

This figure comprises a validation results summary table 700 and an STDP learning efficacy bar chart 702 summarizing all validated claims.

Validation results summary table 700 with columns: Claim | Criterion |

Measured Value | Threshold | Result

Section header: "Phase 7 Control Baseline (EnergyConfig)"

Row A: Closed-loop continuity | All channels 2000 points | 2000 | 2000 | PASS

Row B: Boundedness | All state variables bounded | bounded | bounded | PASS

Row C: Robustness under noise | Std(theta) <= 45 deg | [value] | 45 | PASS

Row D: Input-output gain | corr(L,theta) >= 0.2 | [value] | 0.2 | PASS

Row E: Saturation ratio | sat_theta <= 0.20 | [value] | 0.20 | PASS

Section header: "Phase 7 Full Integration (Control confirms energy depletion)"

Row: Energy at 2000 steps | Energy depleted | -4,697 mWh | < 0 | CONFIRMED

Section header: "MCP Integration (BalancedEnergyConfig)"

**Row: Energy at 2000 steps | Homeostasis at capacity | 100.0 mWh (capped) | > 0 |
CONFIRMED**

Section header: "Phase 8 STDP"

Row F8.1: Weight entropy decrease | final < initial | 2.95 to 2.02 bits | decrease | PASS

Row F8.2: STDP MI > 1.2x frozen | ratio | 6.52x | > 1.20 | PASS

Row F8.3: Late weight delta < 10% early | ratio | 0.00% | < 10% | PASS

STDP learning efficacy bar chart 702 presents the Phase 8 STDP results

**graphically for visual comparison of STDP-enabled versus frozen-weight
conditions.**

Footer: "All claims validated. System is reproducible via git tags."

FIG. 8 — Energy-Bounded Recursive Control Architecture

**This figure is the canonical embodiment and reference signal flow for the
entire system. It shows the complete architecture as a vertically stacked**

block diagram with bidirectional control paths and energy feedback. This is the primary reference figure for patent examination.

System operation is constrained by real-time energy availability derived from physical interaction with the environment, thereby ensuring bounded, non-abstract execution.

Layout — Vertically stacked functional blocks with lateral connections:

TOP — Physical Environment:

The physical environment sits at the top of the diagram.

Two downward arrows from the physical environment:

Left arrow: "(Light / Temp / Force)" — to Sensor Interface 802

Right arrow: "(Mechanical Load)" — to Energy Harvesting 804

TIER 1 — Dual Input (left and right, same level):

Left block:

Sensor Interface 802 receives environmental signals (light, temperature,

force) through specific sensor ICs: BH1750 for light, DS18B20 for

temperature, and piezo input for force. Arrow down from Sensor Interface 802 to Neuromorphic Core 806.

Right block:

Energy Harvesting 804 performs physical energy transduction: piezoelectric

element through full-bridge rectifier to supercapacitor storage. Arrow down from Energy Harvesting 804 to Neuromorphic Core 806.

TIER 2 — Central Processing (full width):

Neuromorphic Core (SNN) 806 occupies the full width of the diagram. It

contains four labeled subsystems:

- "Spiking Neurons" — leaky integrate-and-fire neural population

- "Membrane Dynamics" — $V(t) = V(t-1) \times \text{leak} + I_{\text{adjusted}}$

- "Optional STDP" — spike-timing dependent plasticity (opt-in)

- "Activity-Dependent Energy Cost" — quadratic: $\text{base} + k \times (\text{spikes}/N)^2$

Two downward arrows from Neuromorphic Core 806:

Left arrow: to Recursive Reflection 808

Right arrow: to Energy Governor 810

TIER 3 — Dual Processing (left and right, same level, BIDIRECTIONAL):

Left block:

Recursive Reflection 808 implements the self-observation mechanism. It

feeds aggregate output at timestep t back as input at timestep $t+1$, scaled

by the reflection coefficient. Recursive Reflection 808 maintains prior state memory, output feedback, and temporal context.

Right block:

Energy Governor 810 enforces energy constraints. It monitors storage

level, applies the activity-dependent cost function, and damps activity

when energy is low. Energy Governor 810 manages storage level, cost function, and activity damping.

CRITICAL: Bidirectional arrow between Recursive Reflection 808 and

Energy Governor 810.

Label: "Energy state modulates reflection gain; reflection state

influences energy cost through activity level"

Both Recursive Reflection 808 and Energy Governor 810 have arrows downward to Cognitive Modulation Layer 812.

TIER 4 — Modulation (full width):

Cognitive Modulation Layer 812 occupies the full width of the diagram. It

contains four labeled functions:

- **"Attention Weighting" — prioritizes input channels**
- **"Intention Bias" — goal-directed modulation**
- **"Contextual Suppression / Amplification" — state-dependent gating**
- **"(Optional) External Advisory Input" — external control layer when connected**

Two downward arrows from Cognitive Modulation Layer 812:

Left arrow: to Neural Router 814

Right arrow: to Telemetry & Logging 816

TIER 5 — Output (left and right, same level):

Left block:

Neural Router 814 performs signal arbitration. It routes neural output to

actuators based on priority, gates signals when energy is insufficient,

and prevents output saturation through congestion control. Arrow down from Neural Router 814 to Actuation & Response 818.

Right block:

Telemetry & Logging 816 provides observability output. It records energy

level, system state, and reflection metrics for validation and monitoring. No downward arrow from Telemetry & Logging 816 (terminal output).

TIER 6 — Actuation (full width):

Actuation & Response 818 occupies the full width of the diagram. It

contains servo control, mechanical interaction, and environmental

perturbation subsystems. Actuation & Response 818 drives mechanical

interaction, which perturbs the environment (closing the loop back to

the physical environment at top) and generates mechanical load (feeding

Energy Harvesting 804 at Tier 1).

Two upward feedback arrows from Actuation & Response 818:

1. Arrow back to the physical environment (top): "Environmental Perturbation"

2. Arrow back to Energy Harvesting 804 (Tier 1 right): "Mechanical Load

generates piezoelectric energy"

FEEDBACK LOOPS (mark each distinctly):

Loop A — Energy Harvesting Loop:

Actuation & Response 818 → Mechanical Load → Energy Harvesting 804 →

Neuromorphic Core 806

Style: Bold solid arrows

Label: "ENERGY LOOP — Motor activity generates power for neural computation"

Loop B — Recursive Reflection Loop:

Neuromorphic Core 806 → Recursive Reflection 808 → Neuromorphic Core 806

Style: Dotted arrows

Label: "REFLECTION LOOP — Prior state observation at configurable gain"

Loop C — Environmental Perturbation Loop:

Actuation & Response 818 → Environment → Sensor Interface 802 →

Neuromorphic Core 806

Style: Dashed arrows

Label: "SENSORY LOOP — Actions change environment, environment changes input"

Loop D — Energy Governor Feedback:

Energy Harvesting 804 → Energy Governor 810 ↔ Recursive Reflection 808 →

Cognitive Modulation Layer 812

Style: Thin solid arrows

Label: "REGULATION LOOP — Energy availability constrains all processing"

Key annotations at bottom of figure:

"This architecture is physically grounded: all computation is constrained

by real-time energy availability derived from the system's own mechanical

interaction with its environment."

"No external power source is required for sustained operation under the

balanced energy configuration."

FIG. 8 illustrates one exemplary embodiment; alternative embodiments may

omit, combine, or substitute individual modules while preserving the

energy-bounded recursive control loop.

Sheet numbering: "8/8" at top center

GENERAL DRAWING REQUIREMENTS (per 37 CFR 1.84)

- **Paper size: 8.5 x 11 inches (US Letter)**
- **Top margin: 1 inch**
- **Left margin: 1 inch**
- **Right margin: 5/8 inch**
- **Bottom margin: 3/8 inch**
- **Black lines on white background ONLY**
- **No color, no grayscale shading (use hatching patterns instead)**
- **Line weight: sufficiently heavy for adequate reproduction**
- Each figure labeled: "FIG. 1", "FIG. 2", etc. - Sheet numbers at top center: "1/8", "2/8", etc.
- **No frames or borders around drawing area**
- **All text in figures must be at least 1/8 inch (3.2 mm) high**
- **Reference numerals must be used consistently across all figures**